# What if everyone follows the same best advice?

A new road can worsen traffic equilibrium.
Can more accurate shared advice do the same when people compete for limited opportunities?

## Shared Intelligence, Competing Decisions

A mathematical working paper • Synthetic evidence • Not peer reviewed

THE RESEARCH PATH

# Four papers sharpen the question

| | | |
|---|---|---|
| 1968 / 2005 | **Braess's traffic paradox** | A new route can worsen equilibrium travel times. |
| 2018 | **Informational Braess' paradox** | Learning about routes can hurt affected travelers. |
| 2021 | **Algorithmic monoculture** | A more accurate common ranking can reduce collective hiring performance. |
| 2025 / 2026 | **Strategic hiring** | Congestion-aware differentiation changes the outcome. |

**Our question: when does the private gain from accuracy create a public cost?**

THE MODEL

# Accuracy is only one part of payoff

100 participants • 2 opportunities • Congestion cost per competitor

$$E[W] / N = \mu - c(N - 1)q$$

**μ Expected intrinsic value**

Better advice raises the chance of selecting the good opportunity.

**q Probability two actions coincide**

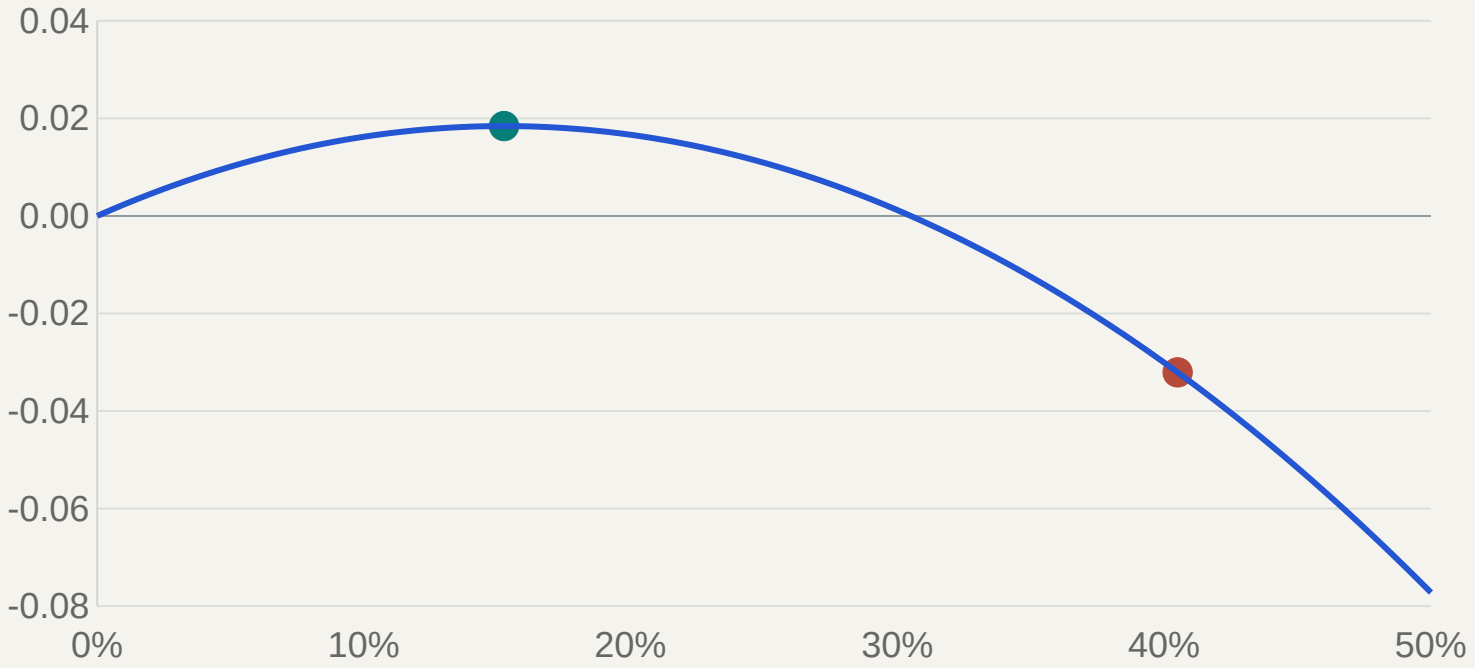
Shared advice raises crowding when it sends people to the same place.

**Private accuracy: 60% • Shared accuracy: 80% • Payoff = value − congestion**

THE RESULT

# Voluntary adoption overshoots the optimum

Payoff change from private-only baseline




**15.3%**

Payoff-maximizing independent adoption

**40.5%**

Equilibrium adoption
Loss: 0.03208 per person

Exact model calculation. The loss is about 0.35% of baseline payoff, not a measured economic loss.

THE REVIEW → THE REFINEMENT

# The result can survive after signals arrive

Each person sees only their own private and shared recommendations.

Illustrative calibration; opponents use the equilibrium strategy.

**WHEN THEY AGREE**

## Follow the recommendation.

Gain from following: 401 / 875 > 0

**WHEN THEY DISAGREE**

## Either action is optimal.

Mixing at p = 40.5% sustains equilibrium.

A scoped equilibrium result. No claim of real-world losses, chaos, or unique equilibrium.

**github.com/triggeredcode/shared-intelligence-competing-decisions**